Project: Worked examples
Title: Rolling bearing life (ISO 281)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the basic and modified rating life of a rolling bearing under a steady load or a duty cycle, with the static safety factor, against required values. It takes ratings, load factors and a_ISO from reference records.

WHAT IT CALCULATES	RELATION
Equivalent load	$F_{dyn} = X_1 \cdot F_r + Y_1 \cdot F_a$ when $F_a/F_r \le e_{lim}$ ($X_1 = 1$, $Y_1 = 0$ for a single-row radial ball bearing), else $F_{dyn} = X \cdot F_r + Y \cdot F_a$ ($Y = 0$ is refused when an axial load acts above e_{lim})
Duty cycle	$F_{dyn} = (\Sigma\ F_{dyn,i}\text{^}p \cdot U_i\ /\ \Sigma\ U_i)\text{^}(1/p)$
Basic rating life	$L_{10} = (C/F_{dyn})\text{^}p$ million revolutions, p = 3 ball, 10/3 roller
Life in hours	$L_{10h} = 10^6 \cdot L_{10} / (60 \cdot n)$
Reliability factor (ISO 281:2007, derived)	$a_1 = 0.95 \cdot [\ln(100/R_{rel})/\ln(100/90)]\text{^}(2/3) + 0.05$
Modified life	$L_{nm} = a_1 \cdot a_{ISO} \cdot L_{10}$
Static safety	$P_0 = \max(X_0 \cdot F_r + Y_0 \cdot F_a, F_r)$, $s_0 = C_0/P_0$

Basis: Dynamic equivalent load and basic rating life[1](cl. 6–7); Modified rating life[1](cl. 9); Static equivalent load and static safety factor[2](cl. 6–7); Rolling-contact bearings[3](Ch. 11, 11th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
type	Bearing type	ball (p = 3)		assumption: Deep groove ball bearing.	any
F_r	Radial load	2.5	kN	assumption: Radial reaction from the shaft calculation.	0 … 1000 kN
F_a	Axial load	0	kN	assumption: Locating bearing elsewhere; no axial load here.	0 … 1000 kN
n	Speed	1500	rpm	spec: Pump datasheet, rated speed	1 … 30000 rpm
C	Basic dynamic load rating	20.3	kN	spec: SKF catalogue, 6206	0.1 … 10000 kN
C_0	Basic static load rating	11.2	kN	spec: SKF catalogue, 6206	0.1 … 10000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
e_{lim}	Load ratio limit e	0.22		spec: Manufacturer factor for single-row deep groove bearings	0 … 5
X_1	Radial factor X_1 for $F_a/F_r \le e$ (1 for a single-row radial ball bearing)	1		default	0 … 5
Y_1	Axial factor Y_1 for $F_a/F_r \le e$ (0 for a single-row radial ball bearing; above 0 for double-row and paired bearings)	0		default	0 … 10
X	Radial factor for F_a/F_r above e	0.56		spec: Manufacturer factor for single-row deep groove bearings	0 … 5
Y	Axial factor for F_a/F_r above e (from the manufacturer; zero makes the axial load count for nothing and is refused when one acts)	2		spec: Manufacturer factor for single-row deep groove bearings	0 … 10
X_0	Static radial factor X_o	0.6		spec: Manufacturer static factor	0 … 5
Y_0	Static axial factor Y_o	0.5		spec: Manufacturer static factor	0 … 10
R_{rel}	Reliability	95	%	assumption: Client reliability target.	90 … 99.9 %
a_{iso}	Life modification factor a_{ISO} (from the manufacturer's tool)	2		assumption: Read from the manufacturer's life tool for clean ISO VG 68 oil.	0.1 … 20
L_{req}	Required life	5000	h	spec: Pump specification, minimum bearing life	1 … 1000000 h
$s_{0,req}$	Required static safety factor	1		assumption: Normal running, smooth loads.	0.5 … 3

3. Assumptions

A1. The ratings, load factors and a_{ISO} are the manufacturer's values, entered as reference records. (No catalogue table ships with TempestOS.)
A2. The reliability factor a_1 uses the ISO 281:2007 Weibull form, 0.64 at 95 % and 0.25 at 99 %. (ISO 281:1990 gave 0.62 and 0.21, and the sheet states the edition.)
A3. Loads are steady within each duty step, and the speed is constant. (Duty steps are weighted by their share of revolutions.)

A4. type (Bearing type) is taken as ball (p = 3). (Deep groove ball bearing.)
A5. F_r (Radial load) is taken as 2.5 kN. (Radial reaction from the shaft calculation.)
A6. F_a (Axial load) is taken as 0 kN. (Locating bearing elsewhere; no axial load here.)
A7. R_{rel} (Reliability) is taken as 95 %. (Client reliability target.)
A8. a_{iso} (Life modification factor a_{ISO} (from the manufacturer's tool)) is taken as 2. (Read from the manufacturer's life tool for clean ISO VG 68 oil.)
A9. $s_{0,req}$ (Required static safety factor) is taken as 1. (Normal running, smooth loads.)

4. Working

[18] $p = 3$ [1](281:2007 cl. 6–7)
$= 3$
Life exponent, ball bearing

[19] $F_{dyn} = X_1 * F_r + Y_1 * F_a$ [1](281:2007 cl. 6–7)
$= 1 \times 2.5\ kN + 0 \times 0\ kN$
$= 2.5\ kN$
Equivalent dynamic load ($F_a/F_r \le e_{lim}$)

[20] $P_0 = max(X_0 * F_r + Y_0 * F_a, F_r)$ [1](76:2006 cl. 6–7)
$= max(0.6 \times 2.5\ kN + 0.5 \times 0\ kN, 2.5\ kN)$
$= 2.5\ kN$
Equivalent static load

[21] $L_{10} = (C / F_{dyn})^p$ [1](281:2007 cl. 6–7)
$= (20.3\ kN / 2.5\ kN)^3$
$= 535.39$
Basic rating life, million revolutions

[22] $L_{10h} = L_{10} * 1000000 * 2 * pi / n$
$= 535.39 \times 1000000 \times 2 \times \pi / 1500\ rpm$
$= 5948.7\ h$
Basic rating life in hours

[23] $a_1 = 0.95 * (ln(1 / R_{rel}) / ln(1 / 0.9))^(2 / 3) + 0.05$ [1](281:2007 cl. 9)
$= 0.95 \times (ln(1 / 95\ \%) / ln(1 / 0.9))^(2 / 3) + 0.05$
$= 0.63791$
Reliability factor (derived, ISO 281:2007 Weibull form)

[24] $L_{nm} = a_1 * a_{iso} * L_{10}$ [1](281:2007 cl. 9)
$= 0.63791 \times 2 \times 535.39$
$= 683.06$
Modified rating life, million revolutions

[25] $L_{nmh} = a_1 * a_{iso} * L_{10h}$
$= 0.63791 \times 2 \times 5948.7\ h$
$= 7589.6\ h$
Modified rating life in hours

[26] $s_0 = C_0 / P_0$ [1](76:2006)
$= 11.2\ kN / 2.5\ kN$
$= 4.48$
Static safety factor

[27] $u_{life} = L_{req} / L_{nmh}$
$= 5000\ h / 7589.6\ h$
$= 0.6588$
Life utilisation (required / modified life)

[28] $u_{s0} = s_{0,req} / s_0$
$= 1 / 4.48$
$= 0.22321$
Static utilisation (required / actual safety factor)

5. Results

SYMBOL	VALUE	DESCRIPTION
L_{nmh}	7590 h	Modified rating life
L_{10h}	5949 h	Basic rating life
F_{dyn}	2.500 kN	Equivalent dynamic load
P_0	2.500 kN	Equivalent static load
s_0	4.48	Static safety factor
L_{10}	535.39	Basic rating life in millions of revolutions
a_1	0.63791	Reliability factor
L_{nm}	683.06	Modified rating life in millions of revolutions
u_{life}	0.66	Life utilisation (governing)
u_{s0}	0.22	Static utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Required life against modified life	5000 h	7590 h	0.66	[1](281:2007 cl. 9)
PASS	Required static safety factor against the static safety factor	1	4.48	0.22	[1](76:2006)

7. Applicability

WARNING records: tabulated values not taken from a reference record are C, C_0 (Bearing load ratings); e_{lim}, X, Y, X_0, Y_0 (Bearing X, Y, e factors); a_{iso} (Bearing aISO life modification factor). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
X_1: not entered, so the default of 1 is used.
Y_1: not entered, so the default of 0 is used.

8. Verification

```
Template mech.bearing-life@1.3.0 Content hash 91324fe50f42db2d Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. International Organization for Standardization. Rolling Bearings—Dynamic Load Ratings and Rating Life. ISO 281. ISO.
2. ISO 76:2006.
3. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.